LLM Week 1

GA 1

A dataset contains four samples that are given below. Each sample is a sequence of words.

- I enjoyed the movie Transformers → 5
- We live among its people now → 6
- They have much to learn → 5
- Freedom is the right of all sentient beings → 8

The sequences have no continuity (that is, changing the order of the sequences does not matter). Let the embedding dimension of each word be $\mathbb{R}^{1\times256}$.

1) Suppose we want the transformer to process each sequence (in its entirety). What should be the minimum context length (window) T? **1 point**

- 5
- 6
- ✓ 8
- 24

we want the transformer to be able to process each sequence in its entireity, meaning the longest sequence determines the min. seq. length. ∴, 8.

Yes, the answer is correct.
Score: 1

Accepted Answers:

8

2) Suppose we pack all the sequences into a batch and pass it to the encoder of the transformer. The dimension of the projection matrices $(W_Q, W_K, W_V) \in \mathbb{R}^{256\times64}$. Then, which of the following tensors denotes the correct dimension of the input to the encoder **1 point**

4 seq, each is padded to max len of 8. Embeed-dim = 256

- $4 \times 256 \times 8$
- ✓ $4 \times 8 \times 256$
- $8 \times 256 \times 64$
- $4 \times 256 \times 64$

$W_Q, W_K, W_V \Rightarrow R^{256\times64}$

input to the encoder is batch of seq. (4, 8, 256)

input (shape) ⇒ {batch size, seq len, embed dim}

Yes, the answer is correct.

3) Suppose we use four attention heads and an additional linear mapping $W_O \in \mathbb{R}^{256\times256}$. Then, how many parameters are there in the multi-head attention sub-layer?

✓ 262144

$W_K, W_Q, W_V \Rightarrow R^{256\times64} \Rightarrow 256\times64 = 16384 \times 3 = 49152 \times 4$ heads

$= 196608$

$W_O \Rightarrow R^{256\times256} \Rightarrow 256^2 = 65,536$

$+ 65536$

262144

Yes, the answer is correct.
Score: 1

Accepted Answers:

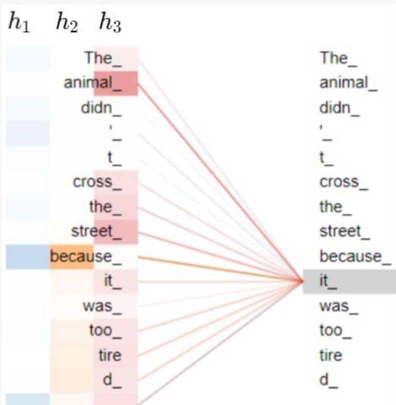
(Type: Numeric) 262144

1 point

Please check the below information and answer the Q4 to Q6

The diagram below shows the attention scores, from three independent heads (h_1, h_2, h_3), for the word ``it_''.

1
2
3
4
5
10
4
15 →



↳ spl. Token ⇒ to show end

$m \Rightarrow$ no. of query tokens (rows)
$n \Rightarrow$ no. of key tokens (columns)
15×15 $(m \times n)$

Each attention head is denoted by a colour (say, $h_3 : red$) and the attention scores computed in the head are represented by its shades (for h_3, shades of red). The darker the colour, the higher the attention score. The last row represents the attention score for a special token. Suppose the size of the attention matrix in $i-th$ head is $A_i \in \mathbb{R}^{m \times n}$. Assume zero-based indexing where required.

4) What is the value of m?

15

Yes, the answer is correct.
Score: 1

Accepted Answers:

(Type: Numeric) 15

1 point

5) What is the value of n?

15

Yes, the answer is correct.
Score: 1

Accepted Answers:

(Type: Numeric) 15

1 point

6) Which of these attention heads captures a strong relation between the words ``it\_" and ``animal_"? *1 point*

- h_1
- h_2
- h_3 $\rightarrow$ very dark red $\Rightarrow$ high score

Yes, the answer is correct.

The input embeddings for the words ``learning", ``brings" and ``joy" are $h_1 = [0.5, 0.25, 1]$, $h_2 = [0.1, 0.25, 0]$, and $h_3 = [0.1, 0.1, 0.9]$, respectively. Note that the embeddings are row vectors. The projection matrices are as follows $W_Q = \begin{bmatrix} 1 & 1 \\ -1 & 1 \\ 0 & 1 \end{bmatrix}$ $W_K = \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$ $W_V = \begin{bmatrix} 0 & 0 \\ -1 & -1 \\ 1 & 1 \end{bmatrix}$ Let z_j denote the linear combination of the value vectors for the $j-th$ word.

7) While constructing z_1, more attention was given to the word ``learning". The statement is *1 point*

- True
- False

Yes, the answer is correct.
Score: 1

Accepted Answers:

True

Index 1:

$$Q_1 = h_1 W_Q^T = [0.5, 0.25, 1]_{1\times 3} \begin{bmatrix} 1 & 1 \\ -1 & 1 \\ 0 & 1 \end{bmatrix}_{3\times 2} = [0.25, 1.5]_{1\times 2}$$

$$K_1 = [0.25, 1.5]$$

$$V_1 = [0.75, 0.75]$$

similarly calculate for all indices.

8) While constructing z_2, all words were given equal attention *1 point*

- True
- False

for $z_1 \Rightarrow$ query is Q_1 & keys are K_1, K_2, K_3

$$Q_1 \cdot K_1 = [0.25, 1.5] \begin{bmatrix} 0.25 \\ 1.5 \end{bmatrix} = 2.3125$$

$$Q_1 \cdot K_2 = 0.2125$$

$$Q_1 \cdot K_3 = 1.525$$

since $Q_1 K_1$ is highest, this proves that 'self' has the highest wts.

$z_2 \Rightarrow$ query is Q_2
& keys are K_1, K_2, K_3
find $Q_2 \cdot K_1$,
$Q_2 \cdot K_2$,
$Q_2 \cdot K_3$.

in 99% of the cases they'll be different.